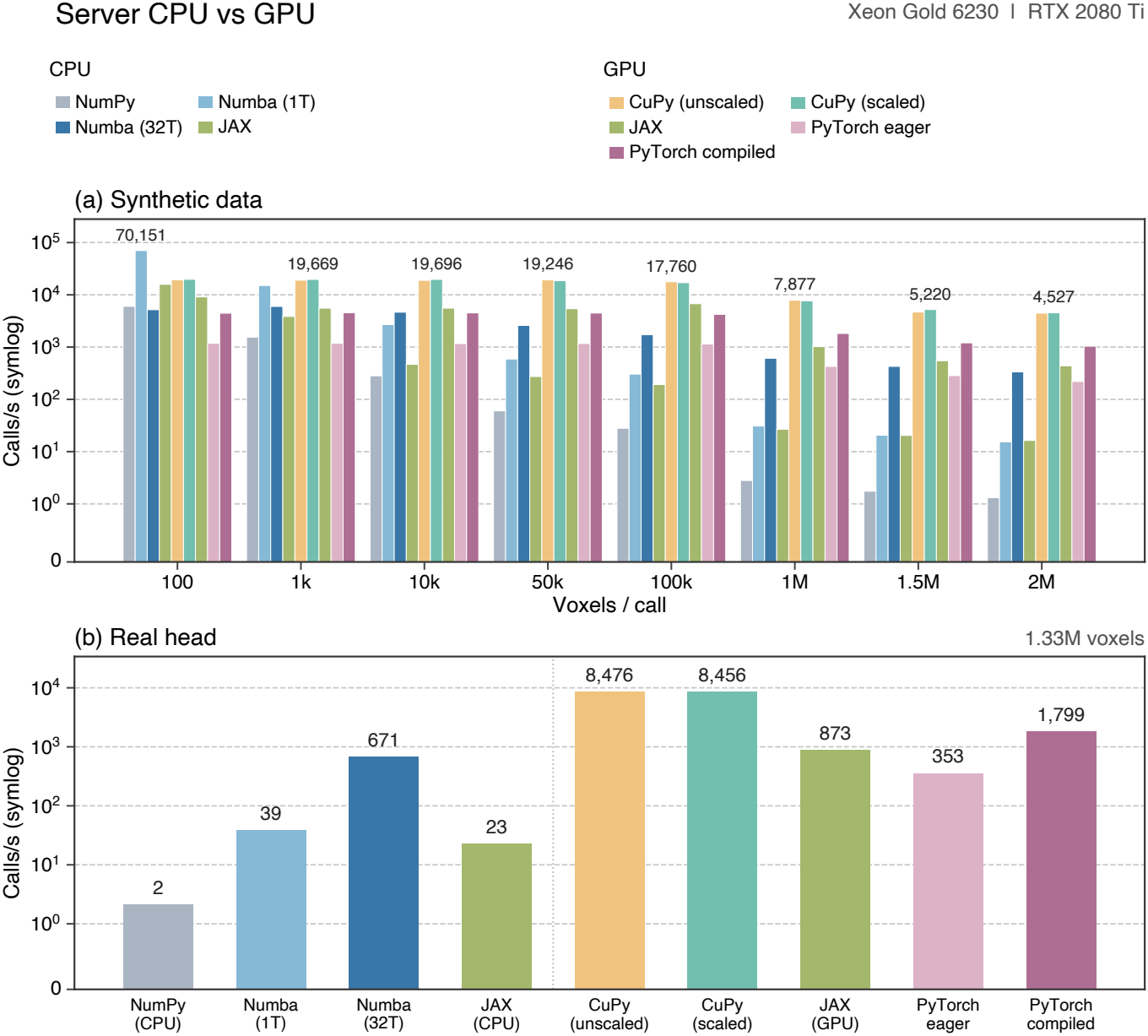
Server CPU vs GPU
Xeon Gold 6230 | RTX 2080 Ti
CPU
NumPy
Numba (1T)
Numba (32T)
JAX
GPU
CuPy (unscaled)
CuPy (scaled)
JAX
PyTorch eager
PyTorch compiled
(a) Synthetic data
Calls/s (symlog)
70,151
19,669
19,696
19,246
17,760
7,877
5,220
4,527
100
1k
10k
50k
100k
1M
1.5M
2M
Voxels / call
(b) Real head
1.33M voxels
2
39
671
23
8,476
8,456
873
353
1,799
NumPy (CPU)
Numba (1T)
Numba (32T)
JAX (CPU)
CuPy (unscaled)
CuPy (scaled)
JAX (GPU)
PyTorch eager
PyTorch compiled